

A REPORT ON

**MATLAB-DRIVEN RISK ASSESSMENT FRAMEWORK TO EVALUATE
CYBERATTACKS ON FOG-ENABLED CRITICAL INFRASTRUCTURE
SYSTEMS**

Submitted by

Preeti

Mentored by

Dr. Abhishek Jain

BML Munjal University

1. INTRODUCTION

Fog computing is an extension of cloud computing where data processing is done closer to the network edge instead of sending everything to a central cloud. This helps reduce delay, improves response time, and supports real-time applications like smart cities, healthcare systems, and industrial automation.

In critical infrastructure systems such as power grids, transportation networks, and communication systems, fog computing plays an important role because these systems require fast and reliable data processing. However, since fog nodes are distributed and connected over networks, they are more exposed to cyberattacks like Denial of Service (DoS), node failures, and attacks on key nodes.

To manage a large number of fog nodes efficiently, clustering techniques are used. In clustering, nodes are grouped together, and a cluster head is selected to manage communication within the cluster and between clusters. This reduces communication overhead and improves network performance.

Different clustering algorithms can be used, such as:

- K-Means clustering (distance-based grouping)
- LEACH (randomized cluster head selection)
- Degree-based clustering (based on connectivity)
- Energy-based clustering (based on node energy)

Each algorithm behaves differently in terms of communication efficiency, energy usage, and robustness under attack.

In this project, a MATLAB-based simulation framework is developed to:

- Deploy a large number of fog nodes
- Form clusters using different algorithms
- Establish communication between nodes based on clusters
- Simulate cyberattacks on the network
- Evaluate Quality of Service (QoS) parameters such as latency, throughput, and packet delivery ratio (PDR)
- Perform risk assessment based on attack impact, vulnerability, and likelihood

The main goal of this work is to understand how different clustering approaches affect network performance and security in fog-enabled critical infrastructure systems, and to identify which approach provides better reliability under cyberattack conditions.

2. LITERATURE REVIEW

Recent research in fog-enabled systems has mainly focused on improving performance and handling large-scale distributed nodes. Many works show that fog architecture is suitable for critical infrastructure because it reduces delay and supports real-time processing. However, when the number of nodes increases, managing communication between them becomes difficult.

To address this, clustering-based approaches are widely used. In most studies, clustering is applied to group nearby nodes and reduce communication overhead. Distance-based methods such as K-Means are commonly used because they form compact clusters and reduce transmission distance. But these methods usually assume ideal conditions and do not consider network failures or attacks.

Another commonly used approach is LEACH, where cluster heads are selected randomly. This helps distribute energy usage across nodes. However, several studies point out that random selection can lead to unstable cluster structures, especially in large networks.

Some researchers have proposed selecting cluster heads based on node properties such as connectivity or remaining energy. Degree-based methods improve communication efficiency by choosing highly connected nodes, while energy-based methods aim to increase network lifetime. These approaches improve specific parameters, but they do not consider how the network behaves under attack conditions.

In terms of security, existing work mainly studies the effect of attacks like Denial of Service and node failures on network performance. It is observed that attacks targeting important nodes, especially cluster heads, can significantly affect throughput and packet delivery.

A key observation from the literature is that most works treat clustering, communication, and security separately. Very few studies combine all three aspects and evaluate them together in a single model. Also, comparative analysis of different clustering techniques under the same attack conditions is not explored in detail.

3. PROBLEM STATEMENT AND GAPS IDENTIFIED

Fog-enabled critical infrastructure systems consist of a large number of distributed nodes that communicate with each other to process and share data. Managing communication efficiently in such networks is a major challenge, especially when the number of nodes is high.

Clustering is commonly used to organize these nodes, but most existing approaches focus only on improving performance parameters such as energy consumption or communication efficiency. They do not consider how the network behaves when it is exposed to cyberattacks.

From the literature, it is observed that clustering methods like K-Means, LEACH, degree-based, and energy-based approaches are usually studied separately. Also, the evaluation is mostly done under normal conditions without considering attack scenarios.

Another important gap is that communication modeling is often simplified. Many studies do not clearly represent how nodes communicate within clusters and between clusters after clustering is performed. This makes it difficult to understand the actual impact of clustering on network behavior.

Security analysis is also limited. Although some works study attacks like Denial of Service or node failures, they do not link these attacks with clustering strategies. In real systems, attacks on cluster heads or critical nodes can have a much higher impact, but this is not properly analyzed.

Based on these observations, the following gaps are identified:

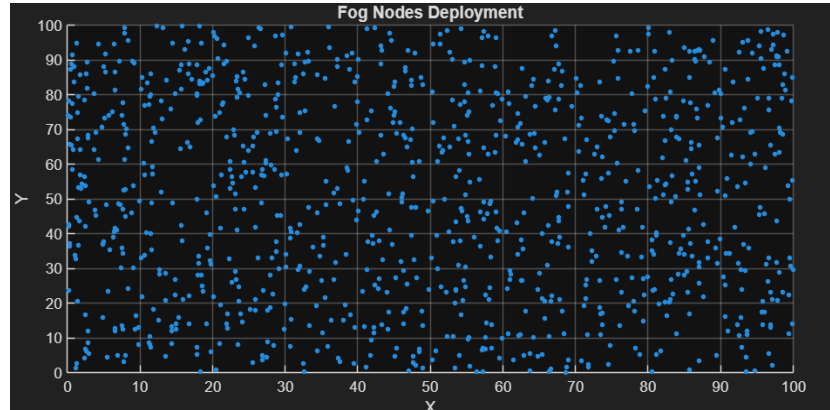
- Lack of a unified framework that combines clustering, communication modeling, and attack simulation
- Limited comparison of multiple clustering algorithms under the same network conditions
- Absence of detailed analysis of how cyberattacks affect clustered fog networks
- No proper risk assessment model that considers impact, vulnerability, and likelihood together
- Insufficient evaluation using multiple QoS parameters such as latency, throughput, and packet delivery ratio

To address these gaps, this project develops a MATLAB-based framework where multiple clustering techniques are implemented, communication is established based on clusters, cyberattacks are simulated, and risk is evaluated using a combined approach.

4. METHODOLOGY

The proposed framework is implemented in MATLAB to simulate a fog-enabled network and evaluate its behavior under cyberattack conditions. The methodology is divided into multiple stages.

4.1 Node Deployment



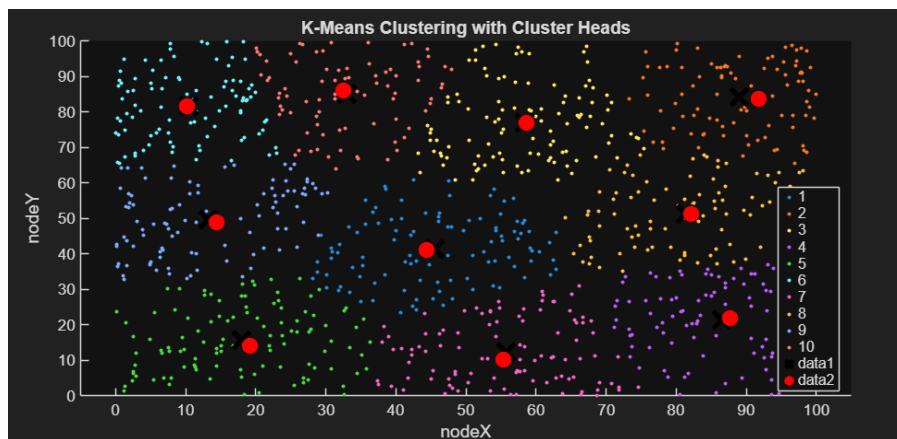
In this step, 1000 fog nodes are randomly deployed in a two-dimensional area of size 100×100 units. Each node represents a fog device in the network.

The deployment is done using random coordinates, so there is no fixed structure or predefined grouping among the nodes. This reflects a practical scenario where devices are distributed over a region without controlled placement.

As shown in the figure, the nodes are spread across the entire area with varying density in different regions. Some areas appear slightly dense while others are relatively sparse, which is expected due to random distribution.

At this stage, no clustering or communication is established. All nodes are independent, and the network does not have any organized structure. This setup serves as the base for applying clustering algorithms in the next step.

4.2 K-Means Clustering and Cluster Head Selection



In this step, clustering is performed using the K-Means algorithm. The deployed nodes are grouped into 10 clusters based on their spatial positions.

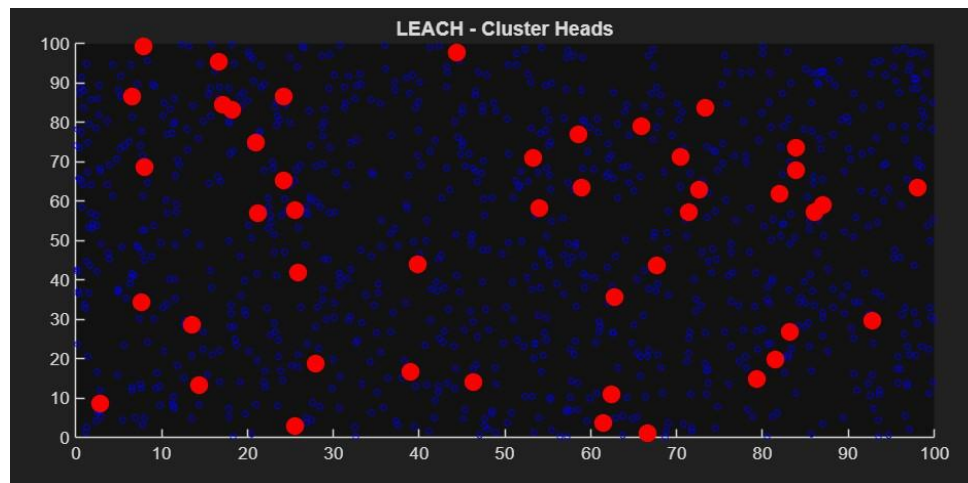
As shown in the figure, each cluster is represented by a different color. Nodes that are close to each other are grouped together, which helps in reducing communication distance within the cluster.

After forming clusters, a cluster head is selected for each cluster. Instead of randomly selecting the cluster head, the node closest to the centroid of the cluster is chosen. These cluster heads are marked with larger red points in the figure.

This method ensures that the selected cluster head is centrally located within the cluster, which helps in efficient communication between nodes and reduces transmission delay.

Compared to the initial deployment, the network is now organized into structured groups. This clustering forms the basis for establishing controlled communication in the next stage.

4.3 LEACH-Based Cluster Head Selection



In this step, cluster heads are selected using the LEACH (Low Energy Adaptive Clustering Hierarchy) approach.

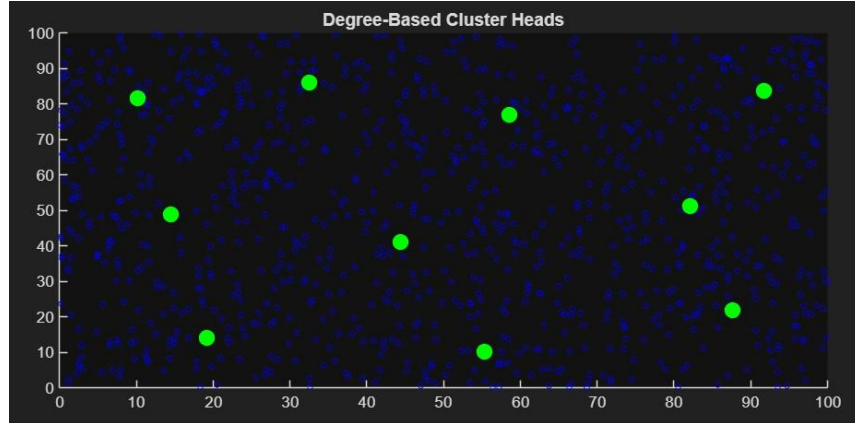
Unlike K-Means, LEACH does not form clusters based on distance. Instead, cluster heads are selected randomly based on a predefined probability. The selected cluster heads are shown as red points in the figure, while the remaining nodes are shown in blue.

As seen in the figure, the cluster heads are distributed across the network without any fixed pattern. Some areas have more cluster heads, while others have fewer. This randomness is a key characteristic of LEACH.

The advantage of this approach is that it helps in distributing the load among different nodes over time. However, since the selection is random, some cluster heads may not be centrally located, which can lead to inefficient communication.

Compared to K-Means clustering, this method does not guarantee balanced or well-structured clusters. This difference becomes important when analyzing communication performance and network behavior under attack conditions.

4.4 Degree-Based Cluster Head Selection



In this step, cluster heads are selected based on node degree, which represents the number of connections a node has with other nodes.

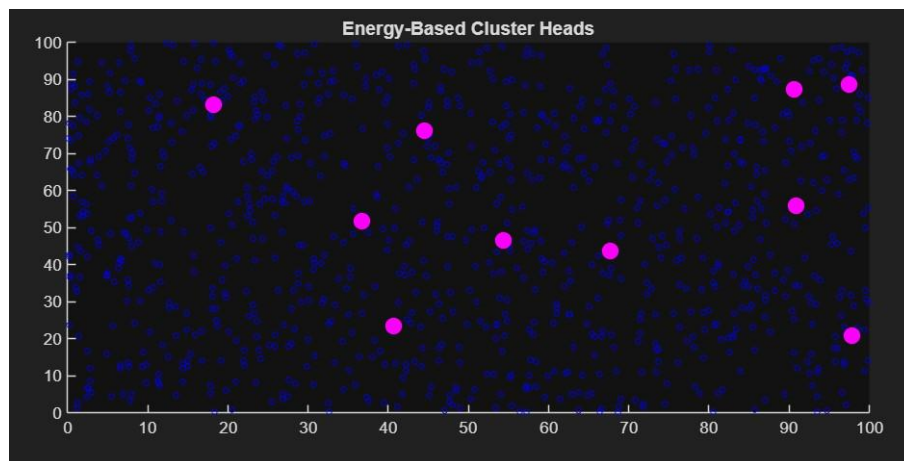
First, communication links between nodes are considered, and the degree of each node is calculated. Nodes with higher degrees are more connected and are therefore selected as cluster heads. These selected nodes are shown as green points in the figure.

As observed, the cluster heads are placed at positions where node density is relatively higher. This is because nodes in dense regions tend to have more connections and higher degree values.

The main advantage of this approach is that highly connected nodes can handle more communication, which improves data flow within the network. However, these nodes may experience higher load, which can lead to faster energy depletion over time.

Compared to LEACH, this method is not random and is based on network structure. Compared to K-Means, it does not consider distance directly but focuses on connectivity.

4.5 Energy-Based Cluster Head Selection



In this step, cluster heads are selected based on the energy levels of the nodes. Each node is assigned a random energy value, and the nodes with the highest energy are chosen as cluster

heads. These nodes are shown as pink markers in the figure, while the remaining nodes are shown in blue.

As observed, the selected cluster heads are distributed across the network but not necessarily in dense regions. This is because selection depends only on energy values and not on position or connectivity.

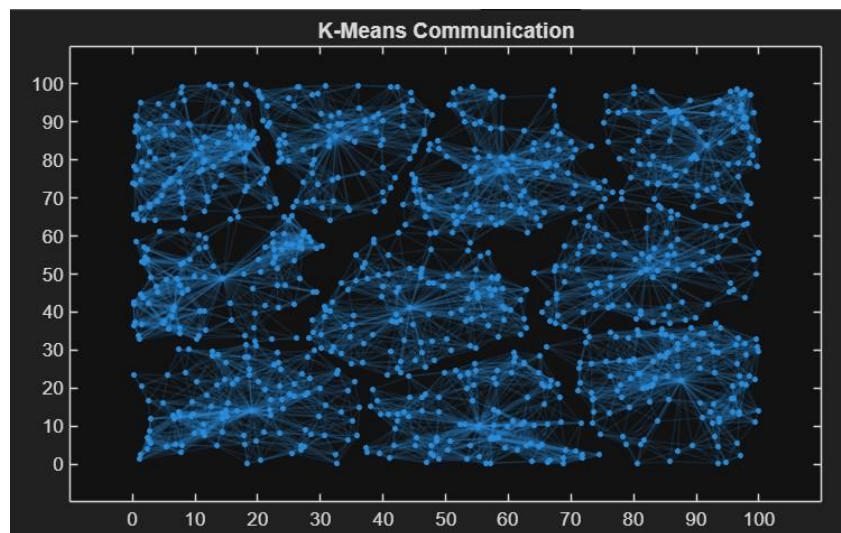
The main advantage of this method is that high-energy nodes can handle communication tasks for a longer time, which helps in increasing network lifetime. However, since location is not considered, some cluster heads may be placed far from other nodes, leading to inefficient communication.

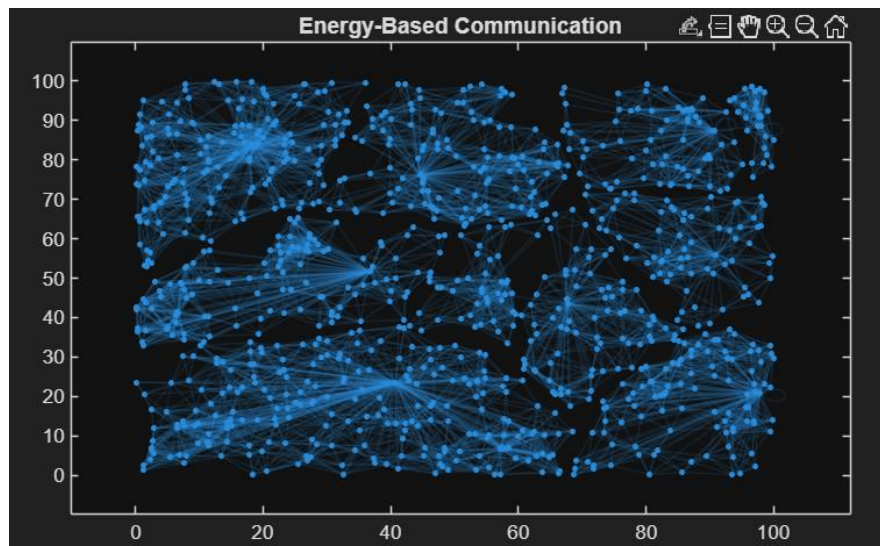
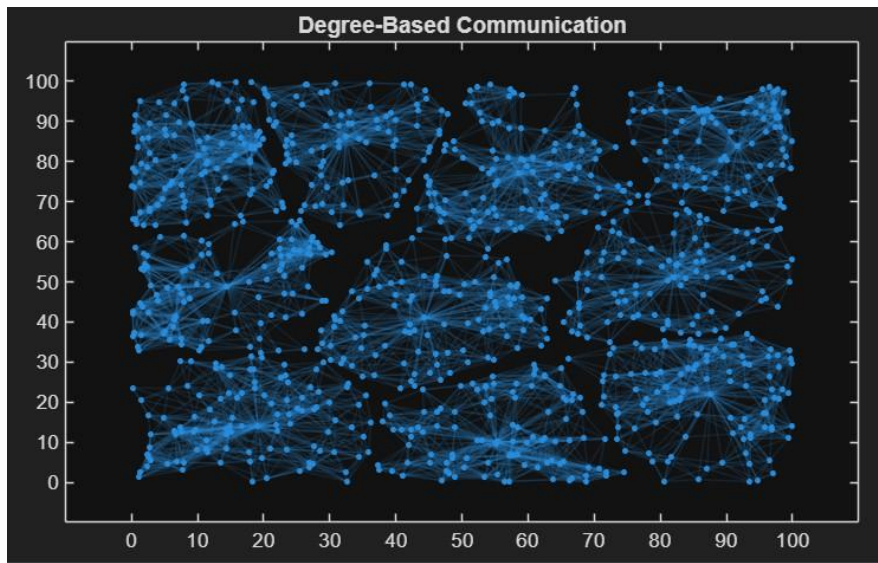
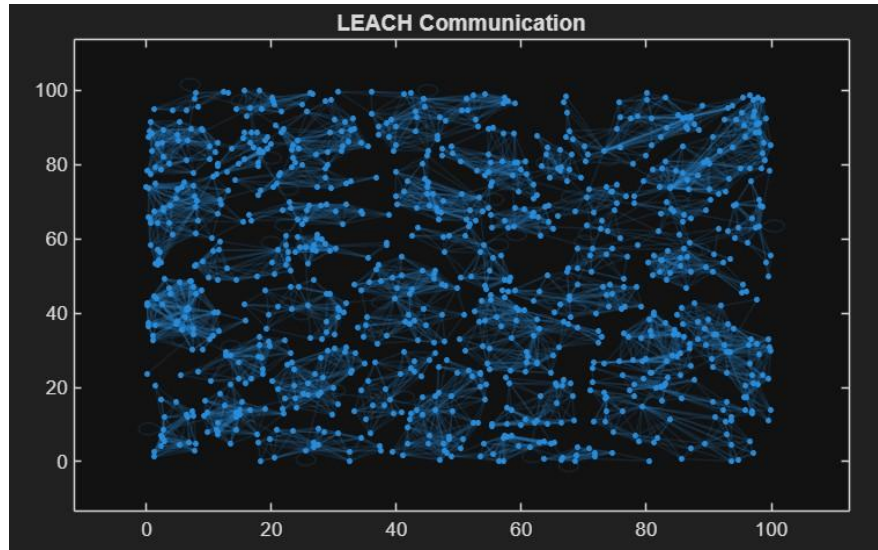
Compared to other methods:

- It is more suitable for energy optimization
- But less effective in terms of communication structure

This difference becomes important when analyzing overall performance and QoS in later stages.

4.6 Cluster-Based Communication





In this step, communication between nodes is established after cluster head selection. Instead of connecting all nodes randomly, a structured communication model is followed.

For all four approaches, the same communication logic is applied:

- Nodes communicate with nearby nodes within a fixed range
- Each node is connected to its cluster head
- Cluster heads communicate with each other

This creates a multi-level communication structure and reduces unnecessary links.

The four communication graphs show how different clustering methods affect the network structure.

In the **K-Means communication graph**, the clusters are clearly visible and well separated. Most connections are within clusters, and there are fewer long-distance links. This results in a balanced and organized communication pattern.

In the **LEACH communication graph**, the structure is more irregular. Since cluster heads are selected randomly, connections are unevenly distributed. Some regions have dense links, while others have sparse communication, leading to less uniform behavior.

In the **degree-based communication graph**, highly connected nodes act as central points. Many connections are concentrated around these nodes, forming hub-like structures. This improves connectivity but can overload certain nodes.

In the **energy-based communication graph**, nodes with higher energy become central communication points. The structure depends on energy distribution rather than position or connectivity. As a result, some long-distance links are observed, and the communication pattern is less compact compared to K-Means.

Overall, K-Means provides the most structured communication, LEACH gives a random distribution, degree-based creates centralized hubs, and energy-based focuses on sustainability rather than communication efficiency.

4.7 Cyberattack Simulation

In order to evaluate the robustness of the network, different types of cyberattacks are simulated in the system. These attacks are introduced artificially to observe their impact on network performance.

Three types of attacks are considered in this work:

1. Denial of Service (DoS) Attack

In this attack, a set of nodes is selected randomly from the network. These nodes behave abnormally and reduce their communication efficiency.

Approximately 10% of the total nodes are affected by this attack. For these nodes:

- Throughput is reduced significantly
- Packet Delivery Ratio (PDR) is also reduced

This simulates congestion or overload conditions in the network.

2. Node Failure Attack

In this case, some nodes are assumed to completely fail and stop functioning. Around 5% of nodes are randomly selected for this attack. For these nodes:

- Throughput is set to zero
- PDR is also set to zero

This represents hardware failure or complete compromise of nodes.

3. Cluster Head Attack

Cluster heads play a critical role in communication, so targeting them has a major impact on the network.

A few cluster heads are selected and attacked. For these nodes:

- Throughput is heavily reduced
- PDR is significantly degraded

This simulates targeted attacks on important nodes in the network.

These attack models help in analyzing how different types of failures and attacks affect the overall performance of the fog network.

4.8 Risk Assessment Model

After simulating the attacks, a risk assessment model is used to evaluate the severity of impact on each node.

The risk for each node is calculated using three main factors:

1. Impact

Impact represents how much the performance of a node is affected due to attacks.

It is calculated based on the reduction in throughput:

$$\text{Impact} = (\text{Throughput Original} - \text{Throughput After}) / \text{Throughput Original}$$

If a node completely fails, the impact becomes high.

2. Vulnerability

Vulnerability indicates how susceptible a node is to attacks.

It is calculated using:

- Node degree (connectivity)
- Remaining energy of the node

Nodes with higher connectivity and lower energy are considered more vulnerable.

3. Likelihood

Likelihood represents the probability of a node being attacked.

Different values are assigned based on attack type:

- DoS attack → medium probability
- Node failure → high probability
- Cluster head attack → very high probability

Nodes not affected by attacks have zero likelihood.

Risk Calculation

The final risk score is calculated as:

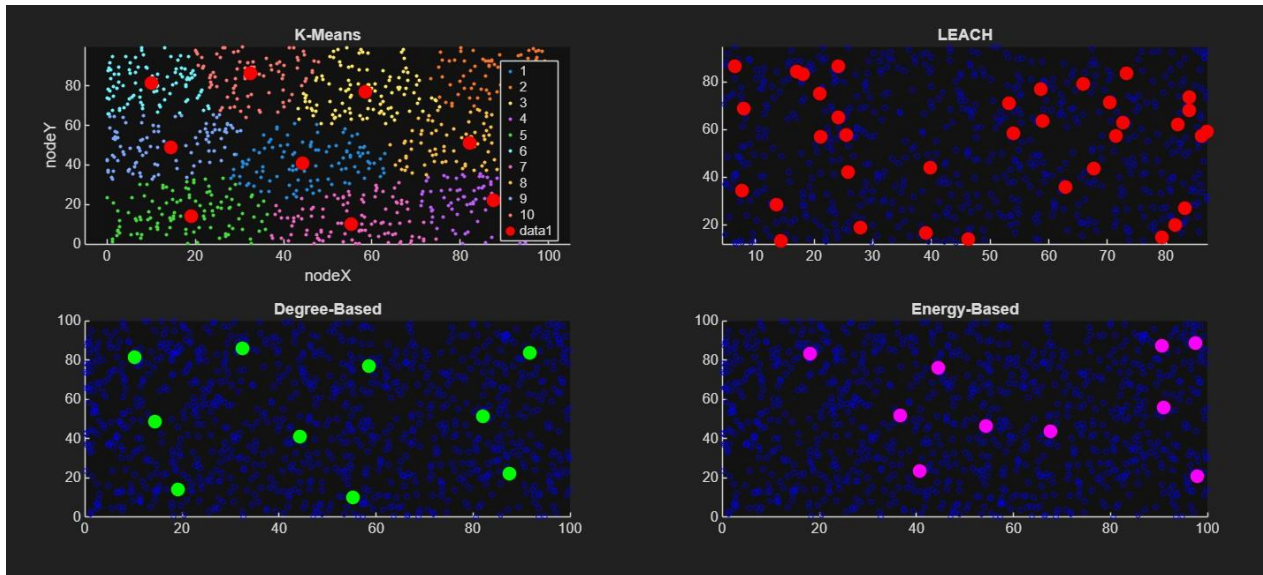
$\text{Risk} = \text{Impact} \times \text{Vulnerability} \times \text{Likelihood}$

The risk values are then normalized between 0 and 1 for better comparison.

This model helps in identifying critical nodes in the network and understanding which parts of the system are more prone to failures or attacks.

5. RESULTS AND DISCUSSION

5.1 Comparison of Cluster Head Distribution



This figure shows the distribution of cluster heads for all four approaches in a single view, making it easier to compare their behavior.

In **K-Means**, nodes are grouped into clear clusters, and cluster heads are placed near the center of each cluster. This results in a balanced and well-organized distribution across the network.

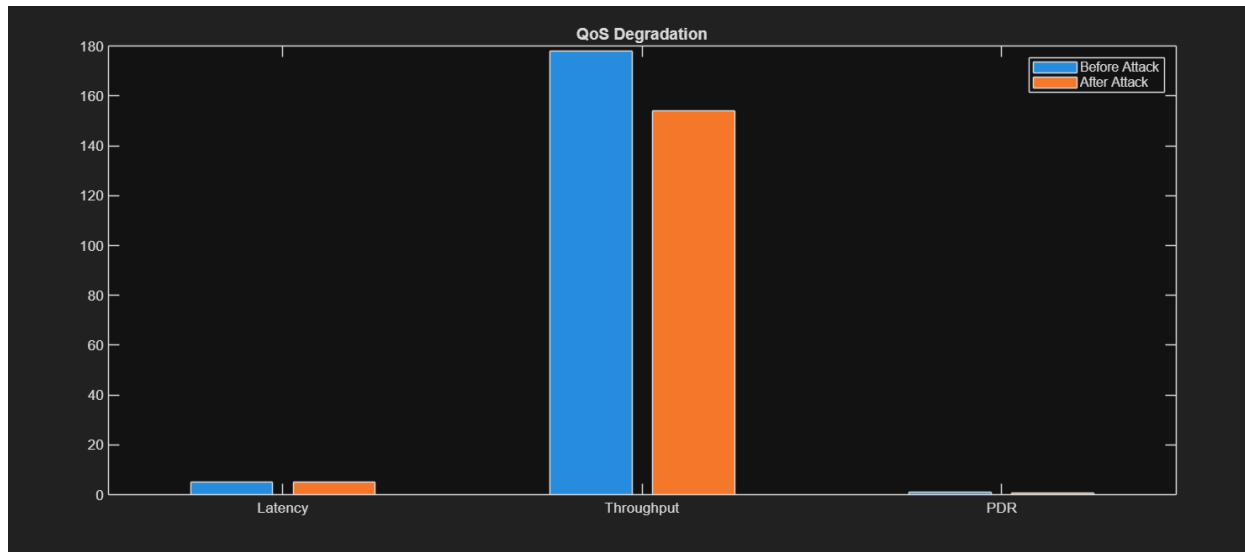
In **LEACH**, cluster heads are selected randomly. As a result, their placement is uneven. Some regions have multiple cluster heads while others have fewer, leading to an unbalanced structure.

In the **degree-based approach**, cluster heads are located in regions where node connectivity is higher. These nodes act as central points in dense areas, which improves communication but may create load concentration.

In the **energy-based approach**, cluster heads are selected based on energy values. Their placement is independent of position or connectivity, so they appear scattered across the network.

Overall, K-Means shows the most uniform distribution, LEACH shows randomness, degree-based focuses on connectivity, and energy-based focuses on node strength.

5.2 QoS Degradation under Cyberattack



This figure shows the comparison of QoS parameters before and after applying cyberattacks on the network.

Three main parameters are considered:

- Latency
- Throughput
- Packet Delivery Ratio (PDR)

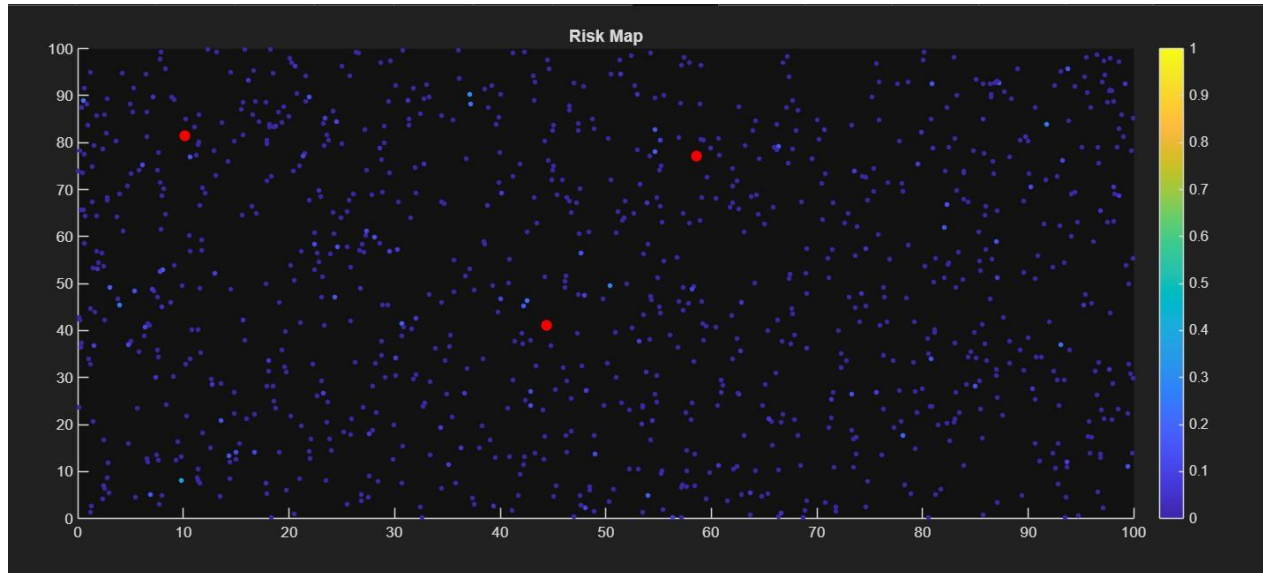
From the graph, it can be observed that **throughput decreases significantly after the attack**. This indicates that fewer data packets are successfully transmitted in the network due to disruptions caused by attacks.

The **packet delivery ratio (PDR)** also reduces after the attack, which shows that the reliability of the network is affected. Some packets are lost due to node failures and targeted attacks.

The **latency shows a slight variation**, which indicates that delay is affected but not as severely as throughput. This is because some communication paths still remain active even after the attack.

Overall, the graph clearly shows that cyberattacks degrade network performance, especially in terms of throughput and packet delivery. This highlights the importance of analyzing network behavior under attack conditions.

5.3 Risk Assessment of Network



This figure shows the risk distribution across all nodes in the network after applying cyberattacks.

The risk value for each node is calculated based on three factors:

- Impact (performance reduction due to attack)
- Vulnerability (based on connectivity and energy)
- Likelihood (probability of attack on the node)

Each node is assigned a risk score, which is normalized between 0 and 1.

In the figure, the color of each node represents its risk level. Nodes with low risk are shown in dark blue, while higher risk nodes appear in lighter colors. The highest risk nodes are highlighted in red.

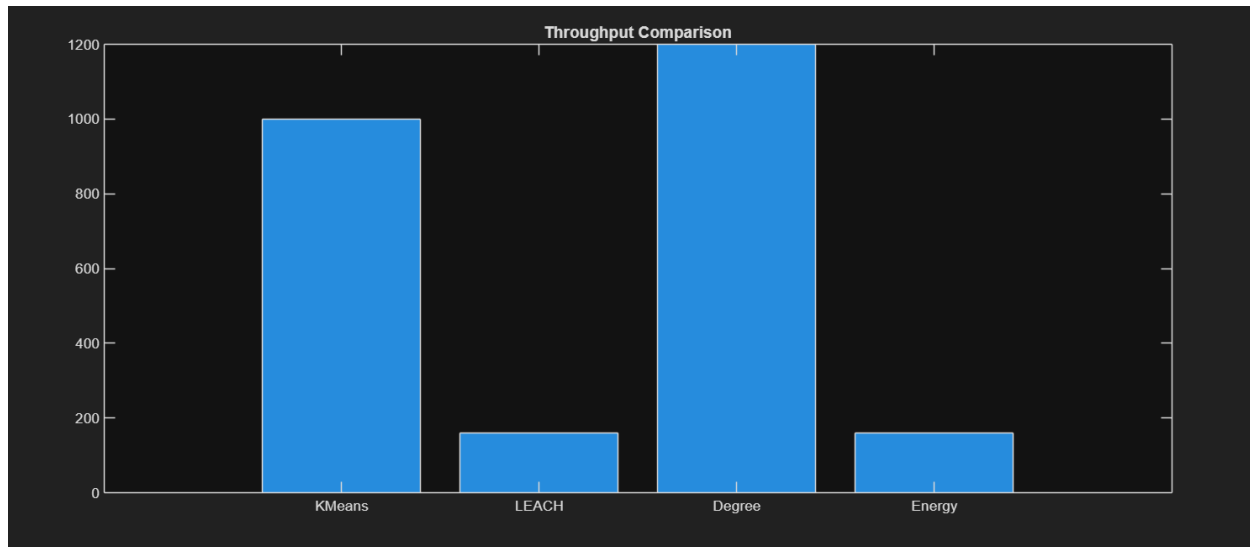
It can be observed that most nodes have low risk values, while only a few nodes show high risk. These high-risk nodes are generally the ones directly affected by attacks or those playing important roles in the network, such as cluster heads.

This visualization helps in identifying critical nodes in the network that require better protection. It also shows how attacks are not uniformly distributed and can significantly affect specific parts of the network.

5.4 Performance Comparison of Clustering Algorithms

This section presents the performance comparison of different clustering algorithms used in the network, namely K-Means, LEACH, Degree-Based, and Energy-Based methods. The comparison is done using key performance metrics such as throughput, latency, energy consumption, packet delivery ratio (PDR), and network lifetime.

5.4.1 Throughput Comparison



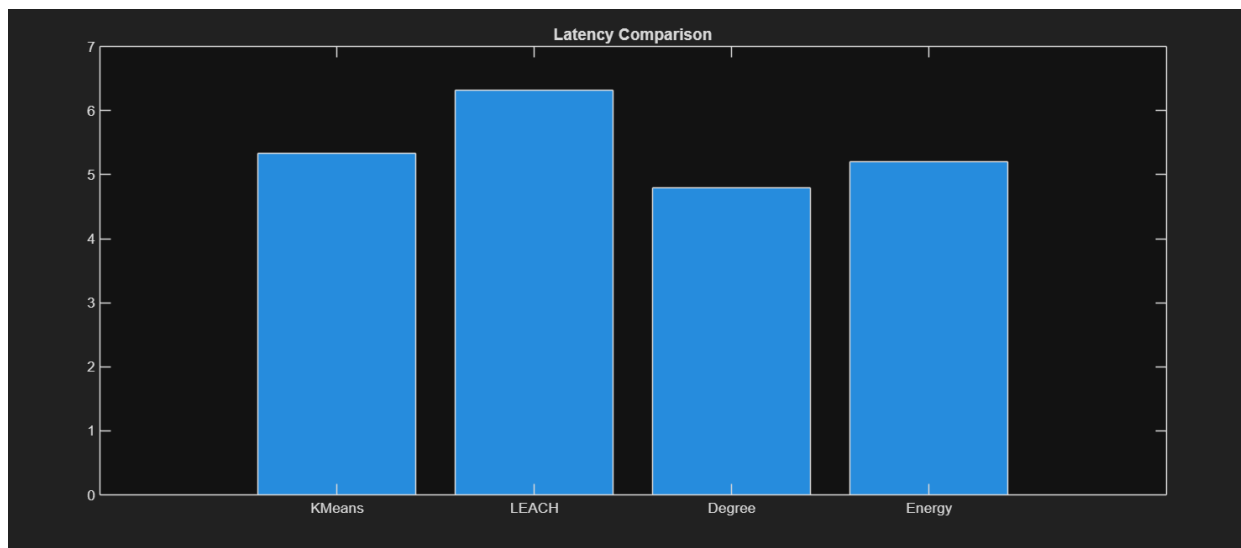
The throughput comparison shows that the **Degree-Based clustering algorithm achieves the highest throughput**, followed by K-Means.

LEACH and Energy-Based methods show significantly lower throughput.

This is because Degree-Based clustering selects nodes with higher connectivity as cluster heads, which improves data transmission efficiency. K-Means also performs well due to distance-based clustering, but it does not always select the most connected nodes.

LEACH performs poorly since cluster heads are selected randomly, which can lead to inefficient communication. Energy-Based clustering focuses on residual energy rather than connectivity, which limits its throughput performance.

5.4.2 Latency Comparison

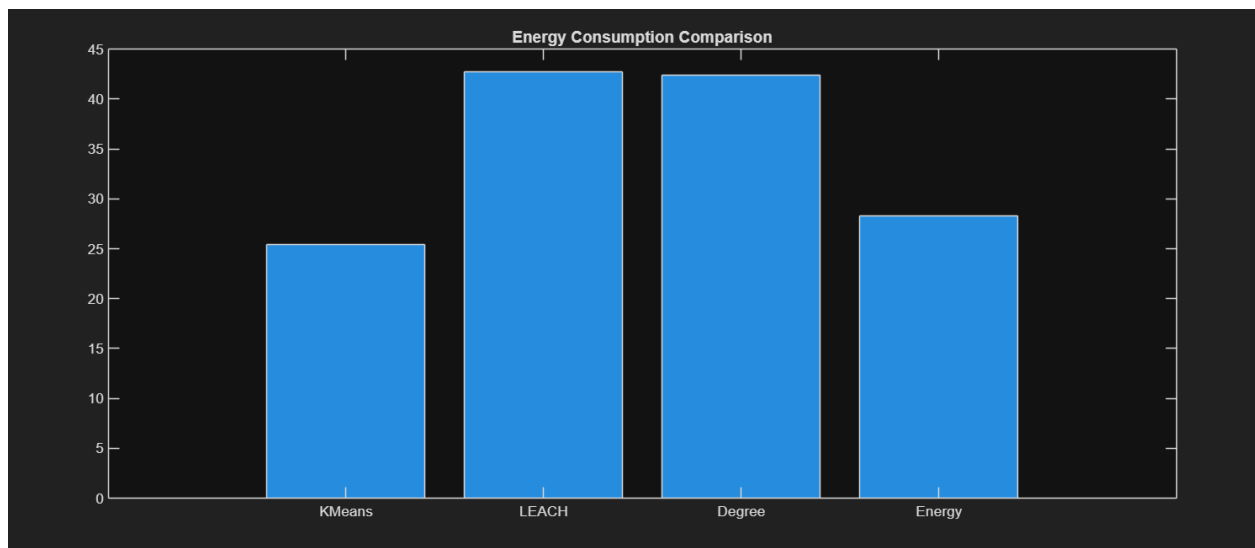


The latency comparison indicates that the **Degree-Based clustering method has the lowest latency**, while LEACH shows the highest delay.

Lower latency in Degree-Based clustering is due to better connectivity between nodes, which reduces communication hops. K-Means and Energy-Based methods show moderate latency performance.

LEACH has higher latency because random cluster head selection can lead to inefficient routing paths and increased delays.

5.4.3 Energy Consumption Comparison



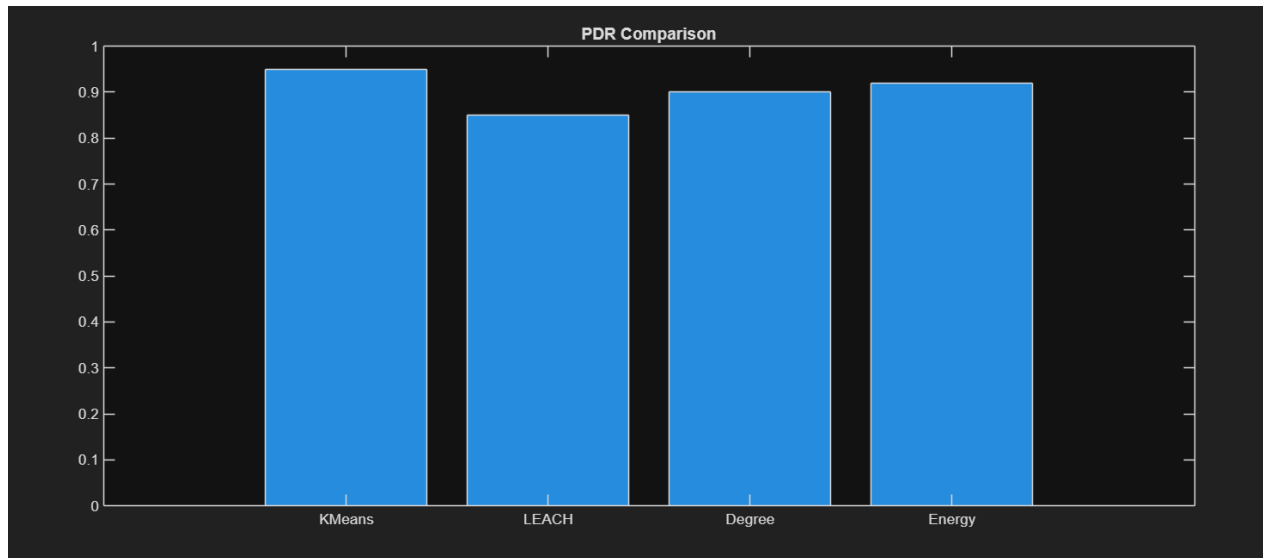
From the energy consumption graph, it is observed that **K-Means clustering consumes the least energy**, followed by Energy-Based clustering.

LEACH and Degree-Based methods consume more energy.

K-Means reduces energy consumption by forming compact clusters based on distance, which minimizes communication overhead. Energy-Based clustering also performs reasonably well since it selects nodes with higher energy levels.

Degree-Based clustering consumes more energy due to high communication load on highly connected nodes. LEACH also shows higher energy usage due to frequent re-selection of cluster heads.

5.4.4 Packet Delivery Ratio (PDR) Comparison



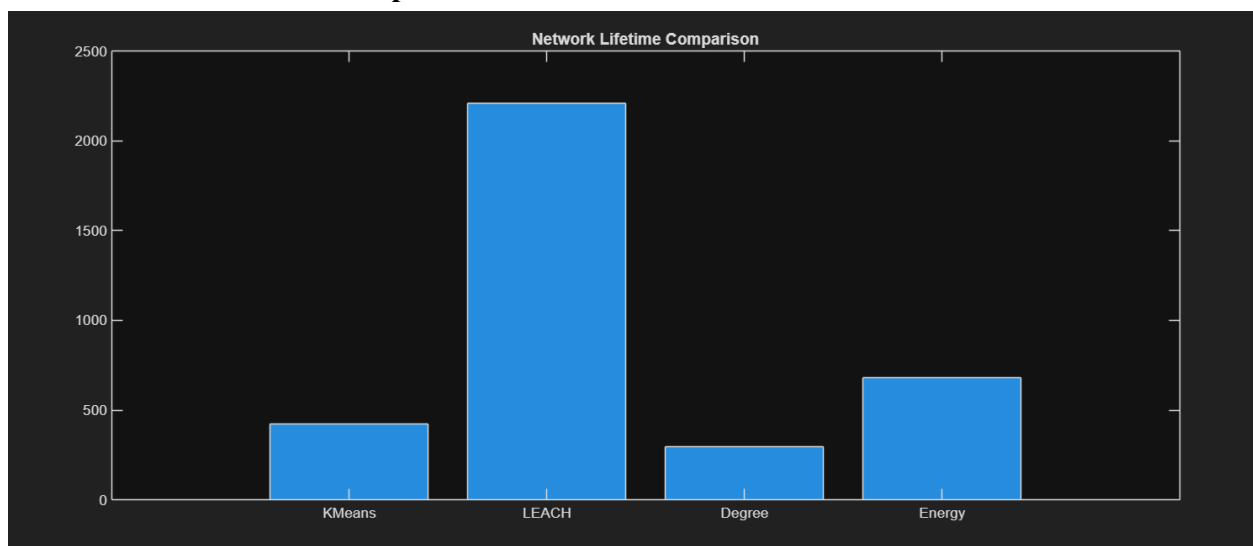
The PDR comparison shows that **K-Means achieves the highest packet delivery ratio**, followed by Energy-Based and Degree-Based methods.

LEACH has the lowest PDR among all algorithms.

Higher PDR in K-Means indicates reliable communication due to well-formed clusters. Energy-Based and Degree-Based methods also maintain good reliability but slightly lower than K-Means.

LEACH performs poorly due to random cluster head selection, which affects consistent packet delivery.

5.4.5 Network Lifetime Comparison



The network lifetime comparison indicates that the **LEACH algorithm provides the longest network lifetime**, followed by Energy-Based clustering.

K-Means and Degree-Based methods show lower network lifetime.

LEACH distributes energy consumption across nodes by rotating cluster heads, which helps in extending network lifetime. Energy-Based clustering also improves lifetime by selecting high-energy nodes.

Degree-Based clustering reduces lifetime because highly connected nodes get overloaded and deplete energy faster. K-Means also shows moderate lifetime due to uneven energy usage among clusters.

Overall Observation

From the above comparisons, it is clear that no single algorithm performs best in all aspects.

- Degree-Based clustering provides better throughput and low latency
- K-Means offers better reliability and lower energy consumption
- LEACH improves network lifetime
- Energy-Based method provides balanced performance

This shows that there is a trade-off between performance, energy efficiency, and network lifetime in fog-based systems.

6. CONCLUSION

In this work, a MATLAB-based framework was developed to evaluate the impact of cyberattacks on fog-enabled critical infrastructure systems. A network of 1000 nodes was deployed, and clustering was performed using four different algorithms: K-Means, LEACH, Degree-Based, and Energy-Based methods.

Cluster-based communication was established, and different types of cyberattacks such as DoS attacks, node failures, and cluster head attacks were simulated. The effect of these attacks was analyzed using QoS parameters including throughput, latency, and packet delivery ratio.

From the results, it was observed that cyberattacks significantly degrade network performance, especially in terms of throughput and packet delivery ratio. The risk assessment model helped in identifying critical nodes in the network, particularly cluster heads, which are more vulnerable to attacks.

The performance comparison of clustering algorithms showed that each method has its own advantages. Degree-Based clustering provides better throughput and lower latency, while K-Means offers better reliability and lower energy consumption. LEACH improves network lifetime, and Energy-Based clustering provides balanced performance.

Overall, the study shows that there is a trade-off between performance, energy efficiency, and network lifetime in fog-based networks. It also highlights the importance of selecting appropriate clustering techniques and securing critical nodes to improve network resilience.